CSC 1800
Organization of Programming Languages
Fall 2025

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Class Meeting: Section 001: **Mendel 290** Tu-Th 10:00am — 11:15am,
Section 002: **Mendel G92** Tu-Th 1:00pm – 2:15pm
Klassner Office Hours: TBA, Online zoom: https://villanova.zoom.us/my/frank.klassner
Course Text: Concepts of Programming Languages, 12th Edition. By Robert W. Sebesta

Course Description:

The course is intended to introduce undergraduate students to the theory behind programming languages' design and to coding practice in various programming paradigms. We will primarily explore the languages C, Java, LISP, Python, as examples of imperative, functional, and interpreted languages, with a discussion of how each supports object-oriented programming. The course's theoretical material will cover syntax, scoping, binding, typing, parameter passing, exception handling, and object-oriented programming, in roughly that order. If time permits, we will include special topics on concurrency. Besides the primary languages the course will draw on features from other languages such as KRpano, Shader Language, Prolog, and JavaScript.

Course Lecture Recordings:

There will be no recordings made of the lecture sessions.

Course Organization and Grading:

The syllabus indicates sections in the text related to the topics in the course. It is important to begin the related readings just before I start leading a topic in class, as my presentation will assume a passing familiarity with the readings. If the reading raises questions for you, please ask them in class. It will make for a more engaging experience – I promise! The purpose of the class presentations on specific programming languages is to examine the languages' features and design, **not** to learn a whole new language in one presentation.

There will be 7-9 semi-weekly graded homework assignments (depending on how we progress in the semester), 4 significant-length programming assignments (2-3 weeks each), one midterm exam, and one comprehensive final exam. Some homeworks will include short programming exercises (< 30 lines of code). Your overall course grade is based on:

30% Final + 25% Midterm + 20% Homework + 25% Programming

The fourth programming project will be a team project. Students will be assigned to teams of two. I will grade team projects as follows. Each project will receive a team grade. This represents the maximum grade anyone on the team can get. After the project is completed, the team members are asked to email me with their assessment of the other team members' effort and intellectual contribution. Using this information, I will assign individual grades based on the team grade. If I receive no email from a team member, I will assume they are satisfied with the other team member's collaborative efforts on the project.

Academic Integrity:

All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. For the College's statement on Academic Integrity, you should consult the *Student Guide to Policies and Procedures*. You may view the University's Academic Integrity Policy and Code, as well as other useful information related to writing papers, at the Academic Integrity Gateway web site: https://library.villanova.edu/research/subject-guides/academicintegrity

As mentioned above, this course will have several programming experiences for students. One will be team-based and all others will be individual. Within a team project all team members are encouraged to share answers and code and help each other out on the source code for the team project. Except for projects explicitly designated as "team projects," **all other graded assignments in the course are individual in scope**. You are **not allowed** to share work, files, etc., on these although you may consult verbally with one another on homework problems if you attribute verbal communication. If what you hand in to me is the product of your own thinking, or is properly attributed, you are maintaining academic integrity. WHEN IN DOUBT, consult me for help.

DEPENDING UPON ITS EGREGIOUSNESS, A VIOLATION OF ACADEMIC INTEGRITY WILL RESULT IN AT LEAST A GRADE OF ZERO FOR THE ASSIGNMENT, UP TO AND INCLUDING A GRADE OF "F" FOR THE COURSE.

AI Policy:

Students are **not permitted** to use ChatGPT to generate or debug code for programming assignments or to generate essays for homework assignments. You may however use it as a tool in a manner similar to using Google or DuckDuckGo to find reference material for learning how to use a new language.

Homework and Program Submission:

Homework assignments are due in BlackBoard by 11:59:59 pm of the day specified. Homework assignments are liable for 5 point-deductions for each day after the due date. Programs are due in Blackboard by 11:59:59 pm the day they are due. Programming assignments will lose 5 points for each day after the due date. All programming assignments will include a written report about the project. The report must be 3-5 pages, line-and-a-half spacing. These can be submitted up to 24 hours after the program's due date. If you are submitting the program late, you need to include both the report and the program when you submit it.

Late Submission/Exam Policy:

Generally, we will follow the policy outlined above. If you anticipate a class absence, notify me as soon as practical about your need to miss class, or to request an extension for an assignment due date. PLEASE NOTE: It is the student's responsibility to notify the instructor if there will be a need to reschedule an exam or assignment. When possible, this should be done at least two weeks in advance of the announced due date.

Covid Mask-Wearing Policy:

Students are welcome, but not required, to wear masks in the course classroom. I will not wear a mask while in front of the class because I feel I am far enough apart from the class for social distancing. Students are welcome, but not required, to wear masks in office consults on campus.

Course Grade:

Using the value obtained from the formula in "Course Organization and Grading," the overall course grade will be assigned based on the absolute scale below. I do not scale any homework, exam, or programming grades. I may raise a student's final letter grade to the next higher "notch" if the student is at the maximum integer value in a notch and has actively and engagingly participated in class discussion. SCORES ARE NOT ROUNDED FOR FINAL GRADE COMPUTATION.

Final Score (out of 100)	**Letter Grade**
95— 100	A
90— 94	A-
87— 89	B+
84— 86	B
80— 83	B-
77— 79	C+

Final Score (out of 100)	**Letter Grade**
74— 76	C
70— 73	C-
67— 69	D+
64— 66	D
60— 63	D-
0 — 59	F

Attendance:

Besides the role of attendance in discretionary final grade bumps (that is, if you're not attending class, it is more difficult to participate in class discussions), there is no official attendance component in the course final grade formula. For this reason, I am not going to manage the "Personal Day" university policy, since my own policy already gives students as many personal days as they need. Although attendance is not mandatory for your grade, please let me know in advance through email as the occasion arises if you will not be attending a given class meeting. This is to help me stay in touch with you and, if necessary, adjust assignment due dates in the case of prolonged absence or personal emergencies. Students who miss a lecture are responsible for obtaining class lecture notes from a colleague. Slides are posted on Blackboard, but these will be the "empty" slides. That is, these will not have anything filled in that was supposed to be filled in by students during the class meeting as part of a learning exercise.

Office of Disabilities and Learning Support Services:

It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. Go to the Learning Support Services website **http://learningsupportservices.villanova.edu** for registration guidelines and instructions. For physical access or temporarily disabling conditions, please contact the Office of Disability Services at 610-519-3209 or 610-519-4095, or email **ods@villanova.edu**. Registration is needed to receive accommodations.

Absences for Religious Holidays:

Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence.
https://www1.villanova.edu/villanova/provost/resources/student/policies/religious holidays.html

Course Topic Schedule (subject to change):

DATE	**TOPIC**	**TEXT/ PROG ASSN.**
Aug 26	Introduction; PL Overview	Ch 1: all;
Aug 28	Project, PL Overview	Ch 1: all; OUT: Java Prog
Sep 02	PL Overview	
Sep 04	Syntax, BNF	Ch 3: 3.1, 3.2
Sep 09	Names, Bindings	Ch 5: all;
Sep 11	Bindings, Scope	
Sep 14	Scope	DUE: Java Prog
Sep 16	C Intro;	OUT: C Prog
Sep 18	Data Types	Ch 6: 6.1 to 6.9, 6.11-6.14
Sep 23	Data Types	
Sep 25	Data Types	
Sep 30	Data Types	
Oct 02	REVIEW	
Oct 07	**Klassner Out in Rome!**	*Midterm*
Oct 09	**Klassner Out in Rome!**	Ideally Due: C Prog
Oct 14/16	***(Fall Break)***	
Oct 21	Python Intro	OUT: Python Prog
Oct 23	Exam Review?	
Oct 28	Expressions & Assignment Statements	Ch 7: 7.1 to 7.7
Oct 30	Expressions & Assignment Statements	
Nov 04	Statement-Level Control Statements	Ch 8: 8.1 to 8.5
Nov 06	**Klassner Out**	
Nov 11	Lisp Intro	Ch 9: 9.1 to 9.6
Nov 13	Subprograms	
Nov 18	Subprograms	Ch 10: 10.1, 10.4
Nov 20	Special Topic?	
Nov 25	Special Topic?	
Nov 27	**Thanksgiving Break**	
Dec 02	Object Orientation	Ch12: 12.1-12.3, 12.4.2, 12.4.4
Dec 04	Object Orientation	
Dec 09	No Class – Friday Schedule	
Dec 11	Review – Final Day of Class	
Dec	**Final Exam** TBA *	

*See https://www1.villanova.edu/university/enrollment-management/registrar/exam-schedules.html for room assignment for the common final exam. Both course sections will take the exam at the same time.

STUDY TIPS AND TIME MANAGEMENT TIPS FOR CSC1800:

This course has a goal of developing students' professional vocabulary. There are many terms defined in the readings and the slides. Students often ask for a list of terms they should study for the midterm or final. To help focus students' study, each homework assignment will have a question that asks students to define a set of terms. These homework terms should serve as the nucleus of your study. Students should be prepared to define these and any other terms on lists in the text or the slides that include these nucleus terms.

Students often ask whether there will be any coding on midterms or finals. I may place questions on tests that ask students to evaluate a snippet of code or comparatively analyze code snippets from different languages, **but I will not use exam questions that ask students to write code from scratch**. Note however that I may ask students to write a few lines from scratch for homework assignment questions.

There are four programming assignments in the course. Even though all four assignments involve solving the same problem, they each will be done in a different language. I will give a brief overview of each language (except for Java) to start the new assignment. It is very important to start working on an assignment as soon as I complete its language's overview. You will find it useful to search online tutorials, including Stack Overflow.